

LEVEL 3 – HARD JEE MCQs

1. Basics of Electrochemistry

1. The molar conductivity of 0.01 M NaCl is $138 \text{ S}\cdot\text{cm}^2\cdot\text{mol}^{-1}$. At infinite dilution, it is $149 \text{ S}\cdot\text{cm}^2\cdot\text{mol}^{-1}$. What is the degree of dissociation?
 - a) 0.074
 - b) 0.086
 - c) 0.092
 - d) 0.105
 2. Which factor has the **least effect** on ionic conductivity?
 - a) Temperature
 - b) Ion size
 - c) Nature of solvent
 - d) Color of solution
 3. A solution shows higher conductivity at 298 K compared to 278 K. Why?
 - a) Increased ion mobility
 - b) Increased concentration
 - c) Reduced viscosity
 - d) Both a and c
 4. The molar conductivity of a weak acid **increases sharply** with dilution. This happens because:
 - a) Ion mobility increases
 - b) Degree of dissociation increases
 - c) Temperature increases
 - d) Both a and b
 5. Conductivity of KCl at 0.01 M is $1.41 \times 10^{-3} \text{ S}\cdot\text{cm}^{-1}$. Find molar conductivity.
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2. Electrolytic and Galvanic Cells

6. For a Zn-Cu cell, $E^\circ_{\text{cell}} = 1.10 \text{ V}$. If $[\text{Zn}^{2+}] = 0.01 \text{ M}$, $[\text{Cu}^{2+}] = 1 \text{ M}$, calculate cell potential at 298 K.
7. In an electrolytic cell, 0.965 C of charge deposits 1 mg of silver. Find electrochemical equivalent.
8. A Daniell cell contains $\text{Zn}^{2+} = 0.1 \text{ M}$ and $\text{Cu}^{2+} = 1 \text{ M}$. Which electrode potential changes faster when the cell discharges?

9. A galvanic cell has Zn/Zn^{2+} and Ag/Ag^+ electrodes. Spontaneous reaction:
- a) $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$
 - b) $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$
 - c) Both
 - d) None
10. Which of the following statements is **true** for electrolytic cells?
- a) Anode is positive
 - b) Cathode is negative
 - c) Non-spontaneous reaction occurs
 - d) All of the above
11. Calculate the quantity of electricity required to deposit 10 g of Cu ($M = 63.5 \text{ g/mol}$, $n = 2$).
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3. Electrode Potential & Nernst Applications

12. Standard electrode potentials: $\text{Zn}^{2+}/\text{Zn} = -0.76 \text{ V}$, $\text{Cu}^{2+}/\text{Cu} = +0.34 \text{ V}$. Find standard EMF of Zn-Cu cell.
13. For the reaction: $2\text{Ag}^+ + \text{Cu} \rightarrow 2\text{Ag} + \text{Cu}^{2+}$, calculate E at $[\text{Ag}^+] = 0.01 \text{ M}$, $[\text{Cu}^{2+}] = 0.1 \text{ M}$.
14. Which of the following metals is the strongest oxidizing agent?
- a) Ag
 - b) Cu
 - c) F_2
 - d) Zn
15. At what concentration of Zn^{2+} will the cell $\text{Zn}/\text{Cu}^{2+}/\text{Cu}$ have $E = 1.00 \text{ V}$ at 25°C ?
16. Nernst equation: $E = E^\circ - (0.059/n) \log Q$. What is the potential of a half-cell if $[\text{Zn}^{2+}] = 0.001 \text{ M}$?
17. A half-cell potential becomes more **negative** when:
- a) Ion concentration increases
 - b) Ion concentration decreases
 - c) Temperature decreases
 - d) None
18. In a Zn-Cu cell, Cu^{2+} concentration is 10^{-3} M . Calculate E_{cell} at 298 K .
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4. Electrochemical Series & Corrosion

19. Metals X, Y, Z have E° values -0.44 , -0.76 , $+0.34$ V. Predict which will corrode fastest.
20. Zn is placed in CuSO_4 solution. Reaction occurs because:
- a) Zn has higher reduction potential
 - b) Zn has lower reduction potential
 - c) Cu is more reactive
 - d) None
21. Corrosion is faster in:
- a) Acidic solution
 - b) Neutral solution
 - c) Basic solution
 - d) Dry conditions
22. Predict if Fe will displace Cu^{2+} from CuSO_4 solution using electrochemical series.
23. A steel bridge is protected using a sacrificial anode made of:
- a) Zinc
 - b) Copper
 - c) Aluminum
 - d) Gold
24. Electrochemical series is used to:
- a) Predict displacement reactions
 - b) Predict corrosion tendency
 - c) Determine oxidizing and reducing agents
 - d) All of the above
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5. Faraday's Laws & Electrolysis Calculations

25. How many grams of Na will be deposited by passing 2 F of electricity?
26. Electrolysis of molten NaCl: products at electrodes are:
- a) Na at cathode, Cl_2 at anode
 - b) Cl_2 at cathode, Na at anode
 - c) NaCl at both electrodes
 - d) NaOH formation
27. 2 F of charge deposits 107 g of Ag. Find the equivalent weight.
28. Calculate the time to deposit 1 g of Cu using a 2 A current.
29. How much H_2 is produced from 18 g of water?
30. Mass deposited in electroplating $\propto$:
- a) Equivalent weight

- b) Current
 - c) Time
 - d) All of the above
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6. Types of Electrochemical Cells & Batteries

31. Lead-acid battery reaction at anode:
- a) $\text{Pb} + \text{SO}_4^{2-} \rightarrow \text{PbSO}_4 + 2\text{e}^-$
 - b) $\text{PbO}_2 + \text{SO}_4^{2-} \rightarrow \text{PbSO}_4$
 - c) Both
 - d) None
32. Hydrogen-oxygen fuel cell produces water and:
- a) Electricity
 - b) Heat only
 - c) Hydrogen
 - d) Oxygen
33. Ni-Cd battery involves:
- a) $\text{Cd} + \text{Ni}(\text{OH})_2 \rightarrow \text{Cd}(\text{OH})_2 + \text{Ni}$
 - b) $\text{Pb} + \text{PbO}_2 \rightarrow \text{PbSO}_4$
 - c) $\text{Li} + \text{CoO}_2 \rightarrow \text{LiCoO}_2$
 - d) None
34. Which battery is preferred in electric vehicles?
- a) Li-ion
 - b) Lead-acid
 - c) Dry cell
 - d) Ni-Cd
35. Rechargeable batteries store energy in:
- a) Chemical form
 - b) Mechanical form
 - c) Thermal form
 - d) None
36. A fuel cell differs from a battery because:
- a) Continuous fuel supply required
 - b) Rechargeable
 - c) Solid electrolyte only
 - d) None

7. Conductometric, Potentiometric & Voltammetric Applications

37. Conductometric titration detects:
- a) Endpoint via conductivity change
 - b) Potential change
 - c) Current change
 - d) Mass change
38. Potentiometric titration can determine:
- a) Strong acid-strong base
 - b) Redox titration
 - c) Both a and b
 - d) None
39. Voltammetry can detect:
- a) Trace metal ions
 - b) Concentration of HCl
 - c) Battery efficiency
 - d) Salt deposition
40. Limiting molar conductivity is:
- a) Extrapolated at zero concentration
 - b) Measured at high concentration
 - c) Measured at 100°C
 - d) None
41. Conductometric titration of weak acid with strong base shows:
- a) Conductivity initially decreases then increases
 - b) Conductivity increases steadily
 - c) Conductivity decreases steadily
 - d) No change

8. Applications of Electrochemistry

42. Electroplating is done to:
- a) Protect from corrosion
 - b) Improve appearance
 - c) Both a and b
 - d) None

43. Electrorefining of copper:
- a) Removes impurities
 - b) Deposits pure Cu on cathode
 - c) Uses Cu anode
 - d) All of the above
44. Chlor-alkali process produces:
- a) NaOH and Cl_2
 - b) NaOH and H_2
 - c) HCl and Na_2CO_3
 - d) None
45. Electrolysis of molten Al_2O_3 produces:
- a) Al at cathode, O_2 at anode
 - b) O_2 at cathode, Al at anode
 - c) Al_2O_3 decomposition only
 - d) None
46. Fuel cells are preferred over batteries because:
- a) Continuous energy generation
 - b) Recyclable fuel
 - c) Less pollution
 - d) All of the above
47. Energy storage using electrochemistry includes:
- a) Lead-acid battery
 - b) Li-ion battery
 - c) Supercapacitors
 - d) All of the above
48. Industrial electrochemistry helps:
- a) Metal extraction
 - b) Electroplating
 - c) Energy conversion
 - d) All of the above
49. Voltammetry measures:
- a) Current vs potential
 - b) Mass vs time
 - c) Conductivity vs concentration
 - d) Voltage vs resistance

50. Potentiometric method uses:

- a) Indicator electrode
- b) Reference electrode
- c) Both a and b
- d) None

✅ **Level 3 total:** ~50–60 MCQs drafted here.

We can continue **Level 3 fully up to ~120–130 questions** next, including **hard numerical + tricky JEE questions** with multi-concept integration, before moving to **Level 4 (HC Verma tough mains questions)**.

Do you want me to continue and complete **Level 3 fully**?

yes

Great! Let's continue **Level 3 – Hard JEE MCQs**, completing the full set (~120–130 questions) with more **numerical and tricky conceptual problems** covering **all topics**.

1. Basics of Electrochemistry (continued)

- 51. The molar conductivity of KCl at infinite dilution is $150 \text{ S}\cdot\text{cm}^2\cdot\text{mol}^{-1}$. If the conductivity of 0.001 M KCl is $0.015 \text{ S}\cdot\text{cm}^{-1}$, calculate the degree of dissociation (α).
 - 52. Conductivity of 0.01 M HCl is $0.014 \text{ S}\cdot\text{cm}^{-1}$, and molar conductivity at infinite dilution is $425 \text{ S}\cdot\text{cm}^2\cdot\text{mol}^{-1}$. Find degree of dissociation.
 - 53. A solution of CH_3COOH shows conductivities of $0.01 \text{ S}\cdot\text{cm}^{-1}$ at 0.01 M. Explain why conductivity increases with dilution.
 - 54. Compare the conductivity of 0.1 M KCl and 0.1 M CH_3COONa . Which is higher and why?
 - 55. Temperature dependence of conductivity is more significant for:
 - a) Strong electrolytes
 - b) Weak electrolytes
 - c) Nonelectrolytes
 - d) All equally
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2. Electrolytic & Galvanic Cells (continued)

56. In electrolysis, 2 F of electricity deposits 31.75 g of Ag. How many grams are deposited by 1 F?
57. A current of 2 A is passed through AgNO_3 solution for 1 hour. Calculate mass of Ag deposited ($M = 107.9 \text{ g/mol}$, $n = 1$, $F = 96500 \text{ C}$).
58. In Zn-Cu galvanic cell, if the solution concentrations are $[\text{Zn}^{2+}] = 0.1 \text{ M}$, $[\text{Cu}^{2+}] = 0.001 \text{ M}$, which electrode potential changes faster?
59. Calculate the EMF of a cell at 298 K if $E^\circ_{\text{cell}} = 1.10 \text{ V}$, $[\text{Zn}^{2+}] = 0.01 \text{ M}$, $[\text{Cu}^{2+}] = 0.1 \text{ M}$.
60. A galvanic cell generates 1 A for 1 hour and deposits 0.65 g of Cu. Find the efficiency of the process if theoretical deposition is 0.70 g.
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3. Electrode Potential & Nernst Equation (continued)

61. $\text{Zn}^{2+}/\text{Zn} = -0.76 \text{ V}$, $\text{Cu}^{2+}/\text{Cu} = +0.34 \text{ V}$. Write the spontaneous reaction.
62. For the cell reaction: $\text{Cu}^{2+} + \text{Fe} \rightarrow \text{Cu} + \text{Fe}^{2+}$, calculate E_{cell} if $[\text{Cu}^{2+}] = 0.01 \text{ M}$, $[\text{Fe}^{2+}] = 0.1 \text{ M}$ at 298 K.
63. If standard electrode potential of $\text{Ag}^+/\text{Ag} = +0.80 \text{ V}$, predict if Ag can oxidize Zn.
64. Nernst equation predicts $E_{\text{cell}} = 0$ when:
- a) $Q = K$
 - b) $Q = 1$
 - c) $Q = 0$
 - d) None
65. Calculate the electrode potential of Zn^{2+}/Zn when $[\text{Zn}^{2+}] = 0.001 \text{ M}$.
66. In Zn-Cu cell, which electrode will act as cathode?
67. At 298 K, $2\text{Ag}^+ + \text{Cu} \rightarrow 2\text{Ag} + \text{Cu}^{2+}$. If $[\text{Ag}^+] = 0.01 \text{ M}$, $[\text{Cu}^{2+}] = 0.1 \text{ M}$, find E_{cell} using Nernst equation.
68. Predict the spontaneity of the reaction using standard potentials: $\text{Fe}^{2+} + \text{Cu} \rightarrow \text{Fe} + \text{Cu}^{2+}$.
69. The more negative the E° of metal:
- a) Stronger reducing agent
 - b) Stronger oxidizing agent
 - c) Less reactive
 - d) Noble metal
70. E_{cell} decreases when:
- a) Reactant concentration decreases

- b) Product concentration increases
 - c) Both a and b
 - d) None
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4. Electrochemical Series & Corrosion (continued)

- 71. Predict which metal corrodes faster: Mg, Zn, Cu, or Ag.
- 72. An iron pipe is immersed in water with Zn anode for protection. Explain which metal is sacrificial.
- 73. Rank metals Na, Al, Fe, Cu according to reactivity using electrochemical series.
- 74. A metal displaces another from its salt solution. Which principle is used?
- 75. Rusting occurs faster in:
 - a) Dry air
 - b) Moist air
 - c) Oil-coated surface
 - d) None
- 76. Galvanic series predicts that:
 - a) Metals higher in series corrode faster
 - b) Metals lower corrode faster
 - c) All metals corrode equally
 - d) None
- 77. $\text{Fe} + \text{Cu}^{2+} \rightarrow \text{Fe}^{2+} + \text{Cu}$ is:
 - a) Spontaneous
 - b) Non-spontaneous
 - c) At equilibrium
 - d) None
- 78. Protection of iron bridge by aluminum rod: which metal is anode?
- 79. Corrosion of metals depends on:
 - a) Electrolyte concentration
 - b) Oxygen availability
 - c) Metal potential
 - d) All of the above
- 80. Metals are coated with:
 - a) Paint
 - b) Zinc

- c) Both a and b
 - d) None
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5. Faraday's Laws & Electrolysis (continued)

- 81. Calculate the mass of Na deposited by passing 1 F of electricity through molten NaCl.
 - 82. 96500 C deposits 107.9 g of Ag. How much mass is deposited by 193000 C?
 - 83. 10 A current passed through CuSO₄ solution for 30 minutes. Mass of Cu deposited?
 - 84. Electrolysis of water generates 22.4 L of H₂ at STP. How much charge passed?
 - 85. A current of 3 A deposits 0.32 g of Ag in 1 hour. Calculate electrochemical equivalent of Ag.
 - 86. Equivalent weight of Cu if 2 F deposits 63.5 g?
 - 87. Electroplating: 5 g of Cu deposited. Current = 2 A. Time required?
 - 88. In electrolysis, $Q = It$. If $I = 5$ A, $t = 2$ hours, calculate Q in C.
 - 89. How many moles of electrons are required to deposit 1 mole of Al (3+)?
 - 90. Electrolytic refining: Impure Cu anode of 100 g deposits how much pure Cu at cathode using 1 F?
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6. Batteries & Fuel Cells (continued)

- 91. Calculate total charge stored in 2 Ah Li-ion battery.
- 92. Lead-acid battery reaction at cathode: $\text{PbO}_2 + \text{H}_2\text{SO}_4 + 2\text{e}^- \rightarrow \text{PbSO}_4 + 2\text{H}_2\text{O}$. Verify electron balance.
- 93. Ni-Cd battery reactions during discharge: $\text{Ni(OH)}_2 + \text{Cd} \rightarrow \text{NiOOH} + \text{Cd(OH)}_2$. Calculate electrons transferred per mole.
- 94. Fuel cell generates 0.5 V per cell. 100 cells in series. Total voltage?
- 95. Why Li-ion batteries are preferred over Ni-Cd for mobile devices?
- 96. Lead-acid battery provides 12 V. Current = 10 A. Energy delivered in 2 hours?
- 97. In H₂-O₂ fuel cell, 1 mole H₂ reacts. Calculate electrical energy produced (1 F = 96500 C, $E = 1.23$ V).
- 98. Which of these can be recharged: Ni-Cd, Li-ion, Lead-acid?
- 99. Electric vehicles prefer:
 - a) Li-ion
 - b) Lead-acid

- c) Dry cell
 - d) Ni-Cd
100. In fuel cell, anode reaction is:
- a) $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$
 - b) $\text{O}_2 + 4\text{e}^- \rightarrow 2\text{O}^{2-}$
 - c) Both
 - d) None
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7. Conductometric, Potentiometric & Voltammetric Applications (continued)

101. Conductometric titration: weak acid + strong base. Endpoint detected when:
- a) Conductivity minimum
 - b) Conductivity maximum
 - c) Zero
 - d) Infinite
102. Potentiometric titration: EMF = 0.05 V at endpoint. Concentration unknown. How determined?
103. Voltammetry detects 10^{-6} M metal ion. Advantage?
104. Molar conductivity at infinite dilution calculated by:
- a) Extrapolation
 - b) Direct measurement
 - c) Standard solution
 - d) None
105. Conductometric titration: 0.1 M HCl with 0.1 M NaOH. Graph shows?
106. Potentiometric method uses:
- a) Reference electrode
 - b) Indicator electrode
 - c) Both
 - d) None
107. In voltammetry, current measured vs:
- a) Applied potential
 - b) Time
 - c) Mass deposited
 - d) Concentration

108. Limiting molar conductivity λ_m^0 is sum of:
- a) Ion contributions
 - b) Electrode potentials
 - c) Current & voltage
 - d) None
109. Conductivity unit in $\text{S}\cdot\text{cm}^{-1}$. Molar conductivity unit:
- a) $\text{S}\cdot\text{cm}^2\cdot\text{mol}^{-1}$
 - b) $\text{S}\cdot\text{m}^2\cdot\text{mol}^{-1}$
 - c) $\Omega^{-1}\cdot\text{m}^{-1}$
 - d) None
110. Conductometric titration useful for:
- a) Weak acid-strong base
 - b) Redox titrations
 - c) Both
 - d) None
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8. Applications of Electrochemistry (continued)

111. Electroplating prevents corrosion because:
- a) Protective coating
 - b) Aesthetic appearance
 - c) Both
 - d) None
112. Electrorefining of copper: cathode weight increases. Which law applied?
113. Chlor-alkali process: produces NaOH from NaCl. Calculate moles NaOH from 2 F of electricity.
114. Electrolysis of molten Al_2O_3 : 1 mole Al requires how many moles of electrons?
115. Energy from fuel cell = 1.23 V, 2 moles of electrons. Calculate energy (J).
116. Electroplating 10 g of Ag. Current = 2 A. Time required?
117. Electrochemical sensors detect:
- a) pH
 - b) Metal ions
 - c) Both
 - d) None
118. Industrial electrochemistry applications:
- a) Metal extraction

b) Electroplating

c) Fuel cells

d) All

119. Voltammetry current-peak proportional to:

a) Concentration

b) Temperature

c) Pressure

d) Time

120. Potentiometric titration useful for:

a) Redox reaction

b) Acid-base reaction

c) Both

d) None